

Alzikrayat Architecture Report

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Course: Advanced Web Technologies

Project: Alzikrayat Photo Sharing Application

1. Project Overview

Alzikrayat is a photo-sharing web application built with PHP and MySQL. Users can register, log in, upload photos with descriptions, browse a gallery, view photo details, delete their own photos, and add comments.

The project demonstrates MVC architecture, three-tier separation, Manual Routing, database integration, authentication, sessions, cookies, validation, and responsive design. It was developed without Laravel, Django, Spring, CMS, or ORM tools.

2. Architecture & Project Structure

The application uses three layers. The **Presentation Layer** contains PHP views, HTML, Bootstrap, CSS, and JavaScript. The **Application Layer** contains Controllers and the custom Router. The **Data Layer** contains Models and the PDO database connection.

`public/index.php` is the front controller. `core/Router.php` matches routes with Regular Expressions and dispatches dynamic IDs, such as:

Plain Text

```
GET  /photos
GET  /photo/12
POST /photo/store
POST /photo/12/comment
POST /photo/12/delete
```

The project structure is:

Plain Text

<code>config/</code>	Database configuration
<code>core/</code>	Model, Router, and bootstrap files
<code>controllers/</code>	Authentication, photo, and comment logic
<code>models/</code>	User, Photo, and Comment models
<code>views/</code>	Authentication, home, layout, and photo pages
<code>public/</code>	Entry point, CSS, and uploaded images
<code>database.sql</code>	MySQL schema

3. Database Design

The database contains three normalized tables: `users` , `photos` , and `comments` .

The `users` table stores account information and hashed passwords. The `photos` table stores image metadata and connects every photo to its owner. The `comments` table stores comments and connects them to users and photos.

Foreign keys maintain referential integrity, and cascade deletion removes related records when a parent user or photo is deleted. PDO prepared statements are used for database queries to reduce SQL Injection risks.

4. Main Features

The main features are:

- User registration, login, logout, and session management.
- Secure password hashing and a seven-day last-login cookie.
- Photo upload with file type and size validation.
- Responsive gallery with three-column, four-column, and list layouts.
- Photo details with title, author, description, date, and comments.
- Deletion restricted to the photo owner.
- Comments restricted to authenticated users.
- CSRF protection, server-side validation, prepared statements, and escaped output.

5. Novelty/Advanced Features

The project includes multiple gallery layouts, a custom dark-blue and gold visual design, responsive Bootstrap components, a last-login cookie, and a JavaScript show/hide password button.

It also applies ownership validation when deleting photos. A user cannot delete another user's photo by changing the ID in the URL because the server checks the current user's ID before deletion.

6. Conclusion

Alzikrayat is a complete photo-sharing application that demonstrates MVC, Manual Routing, MySQL relationships, authentication, CRUD operations, file uploads, comments, validation, security, and responsive design.

The application separates its presentation, application, and data responsibilities and provides a clear foundation for future features such as search, tags, favorites, pagination, and image filters.